

17/04/2026

RAMPS: Commissioning & Science Run 1

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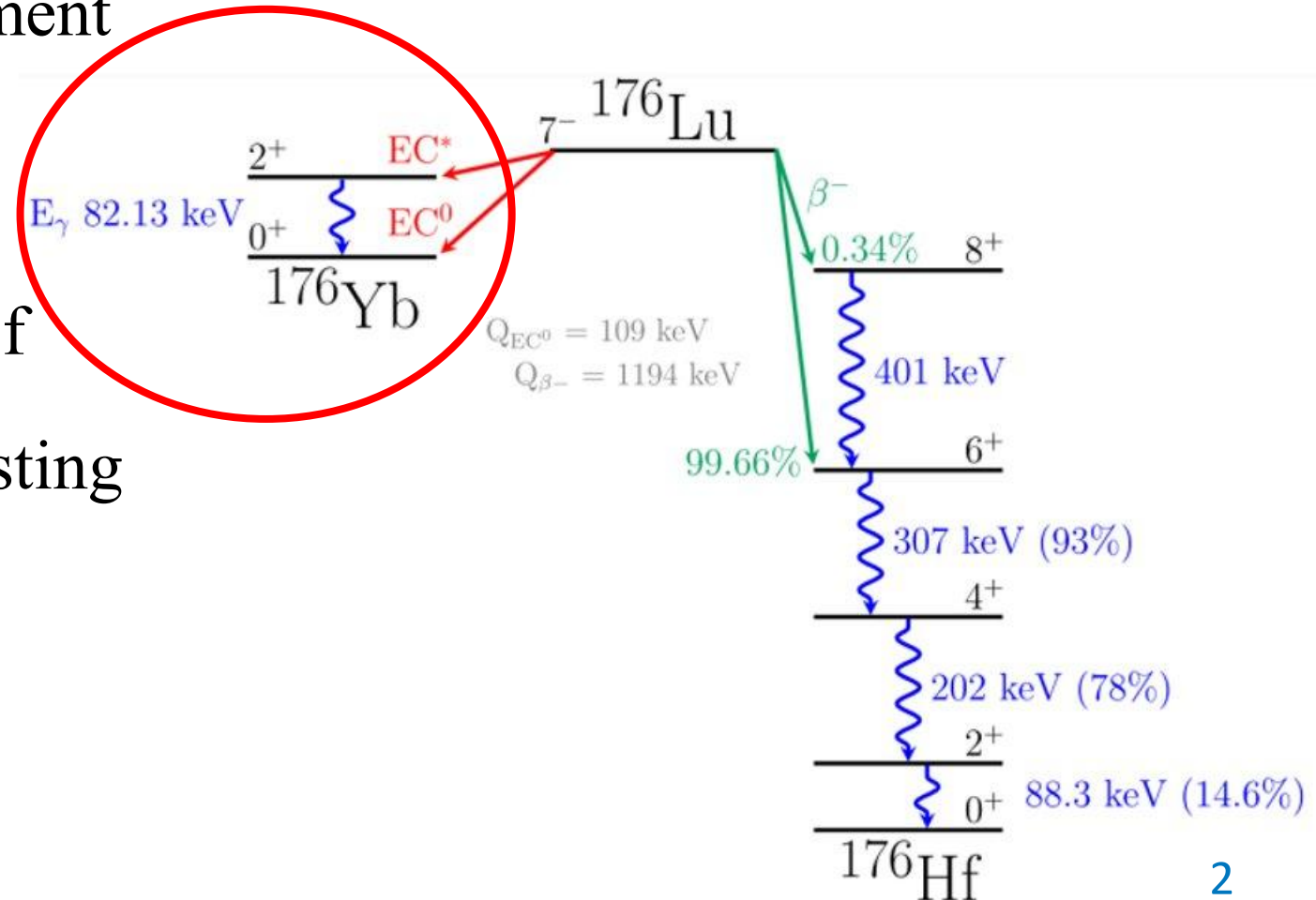


What is RAMPS?

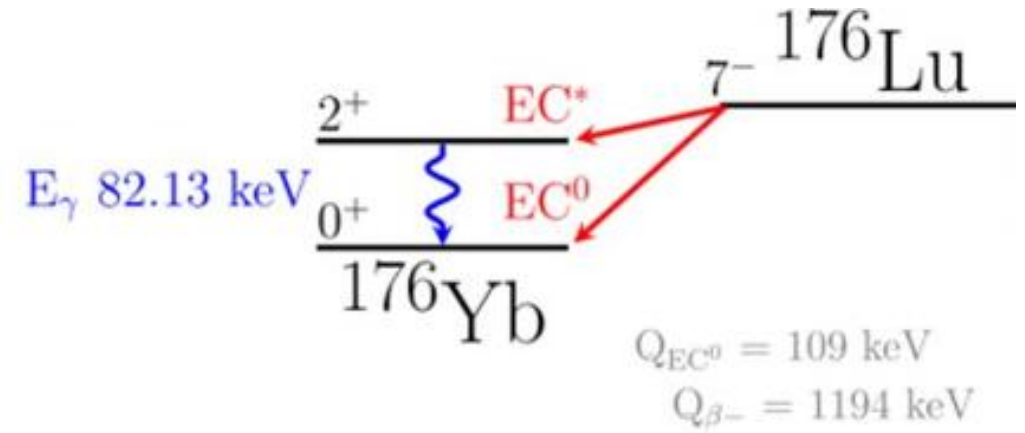
RadioActive isotope Measurement Program at SNOLAB

Performing novel, precision or standardization measurements of long lived (highly forbidden) isotopes using or enhancing existing equipment at SNOLAB

- Geochronology
- Astrophysics
- Nuclear physics



^{176}Lu Decay



- ^{176}Lu has a fifth forbidden EC^* decay
- Current experimental lower limits indicate the half life at 10^{13} - 10^{14} y
- Current theoretical limits: $>1 \times 10^{19}$ y
- LYSO Crystal Source

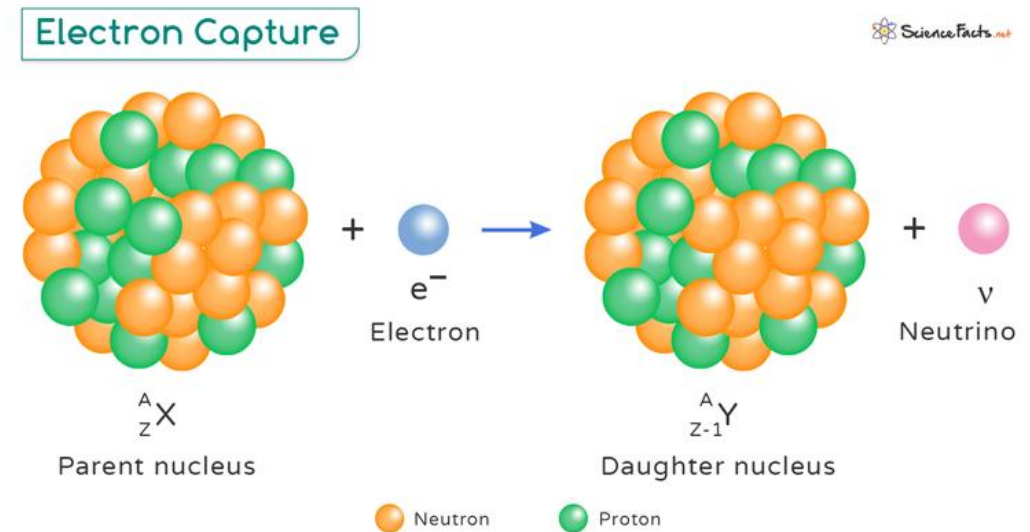
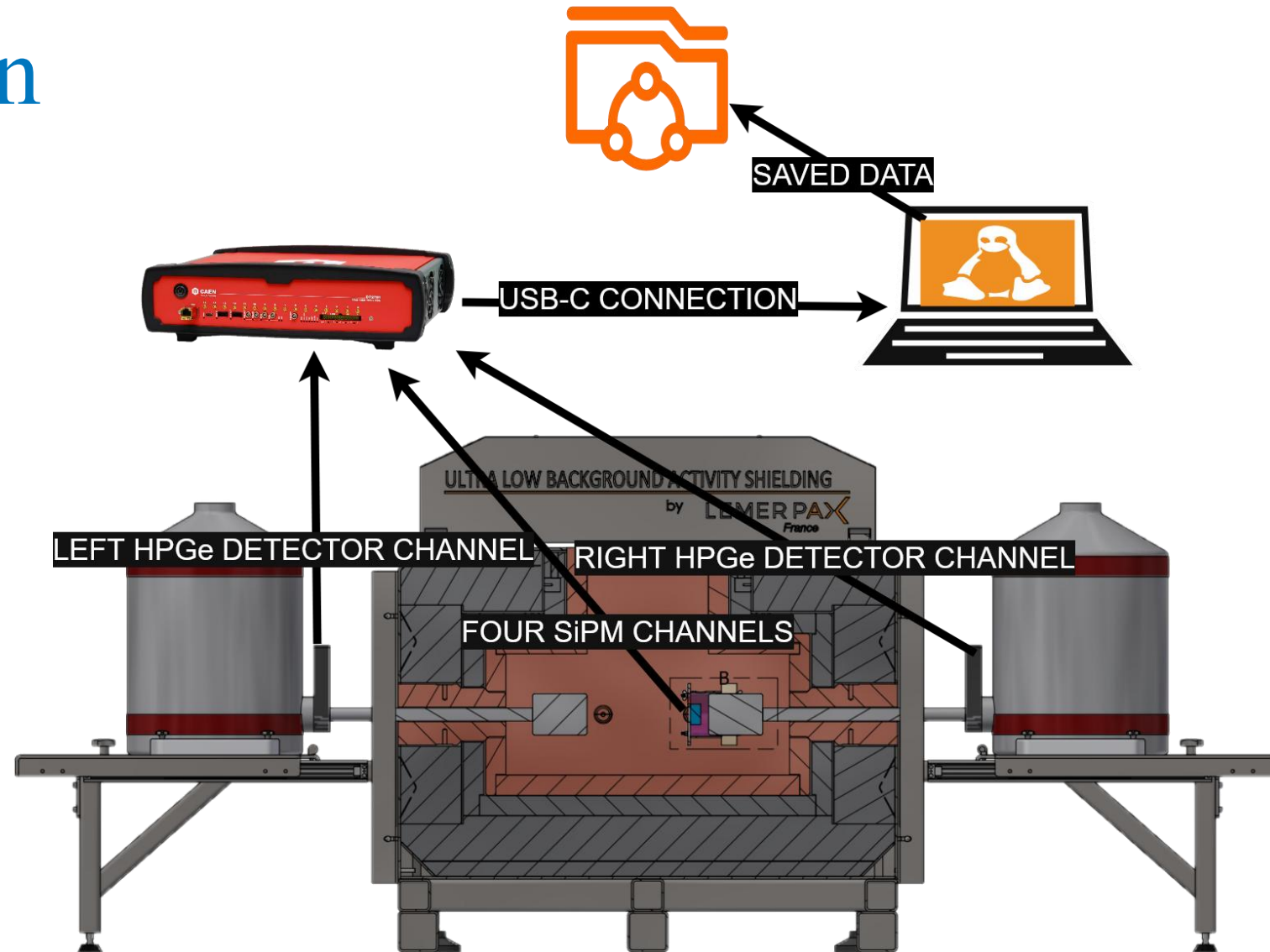


Figure 2: Electron capture process diagram (Source: *Science Facts*)

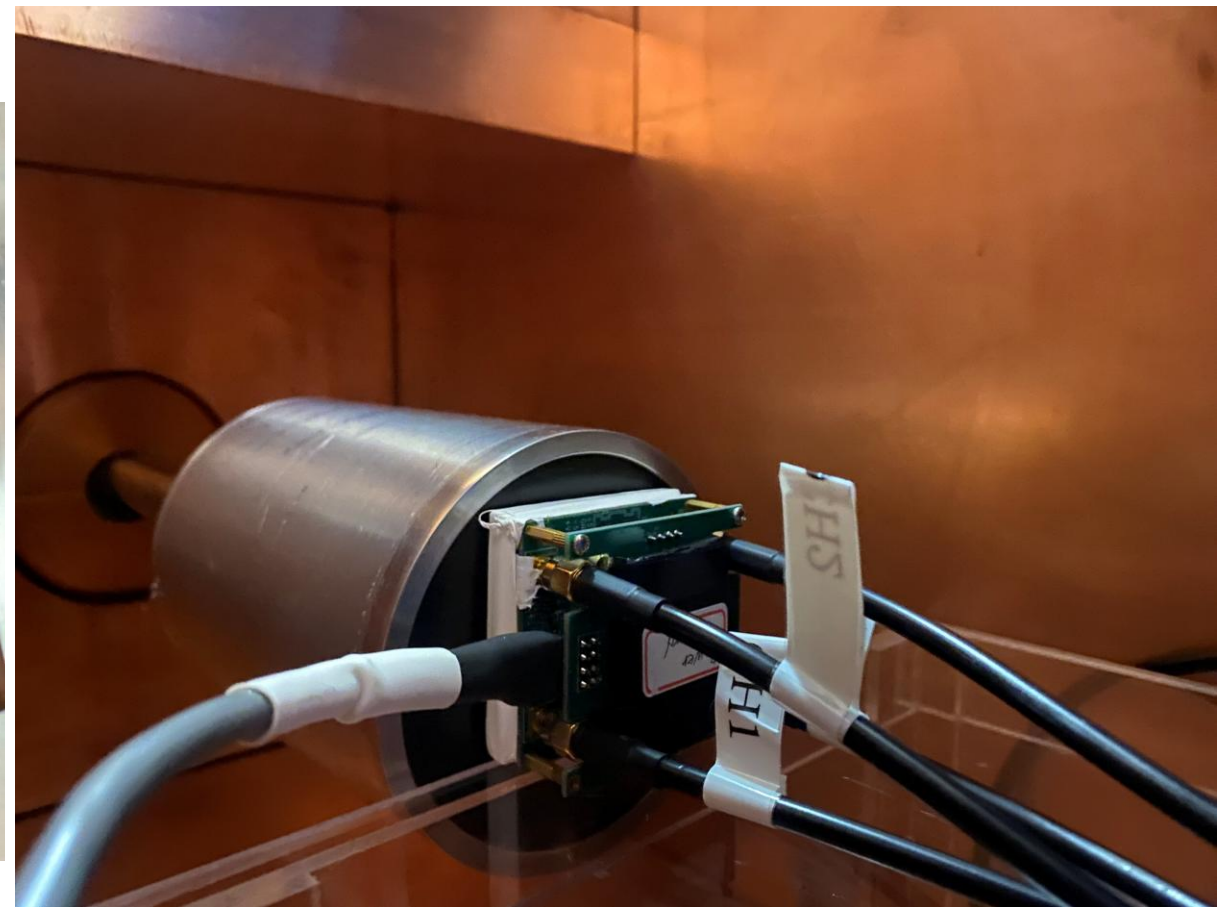
Experiment Design

Dual Coincidence Design

1. CTBT High Purity Germanium Detector (Health Canada)
2. Silicon Photomultiplier (Canadian Nuclear Laboratories)

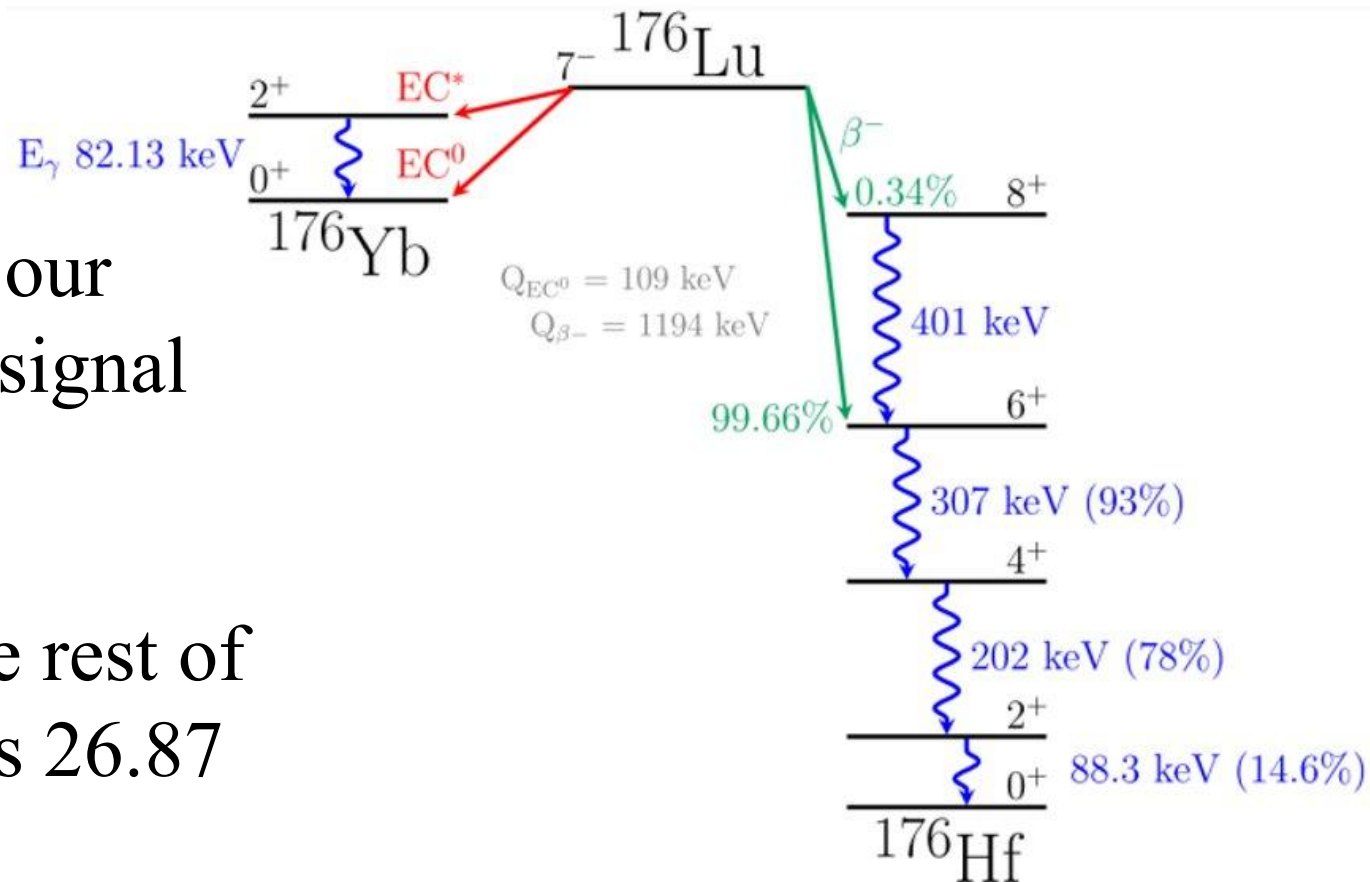


Experiment Design



27keV cut

- When measuring with purely our HPGe, the β^- overpowers our signal
- How to remove?
- EC Q-value = 109 keV
- γ ray is 82.13 keV, leaving the rest of the energy to be scintillated as 26.87 keV light
- Coincidence helps us with this!



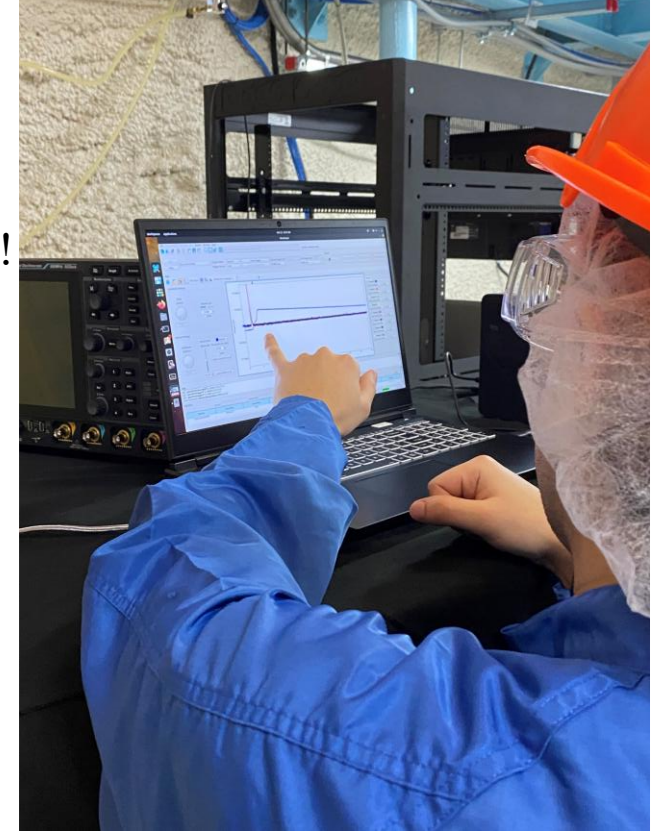
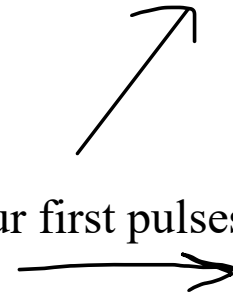
$$Q_{\text{EC}^*} = 109 \text{ keV} - 82.13 \text{ keV} = 26.87 \text{ keV}$$

Science Run 1

- Over four days, we took over 20 Tb of waveform data
- Calibration data with Eu152, Fe55, Ba133
- Worked out very well, ran overnight, with no dead time
- Over 30 hours of science data!

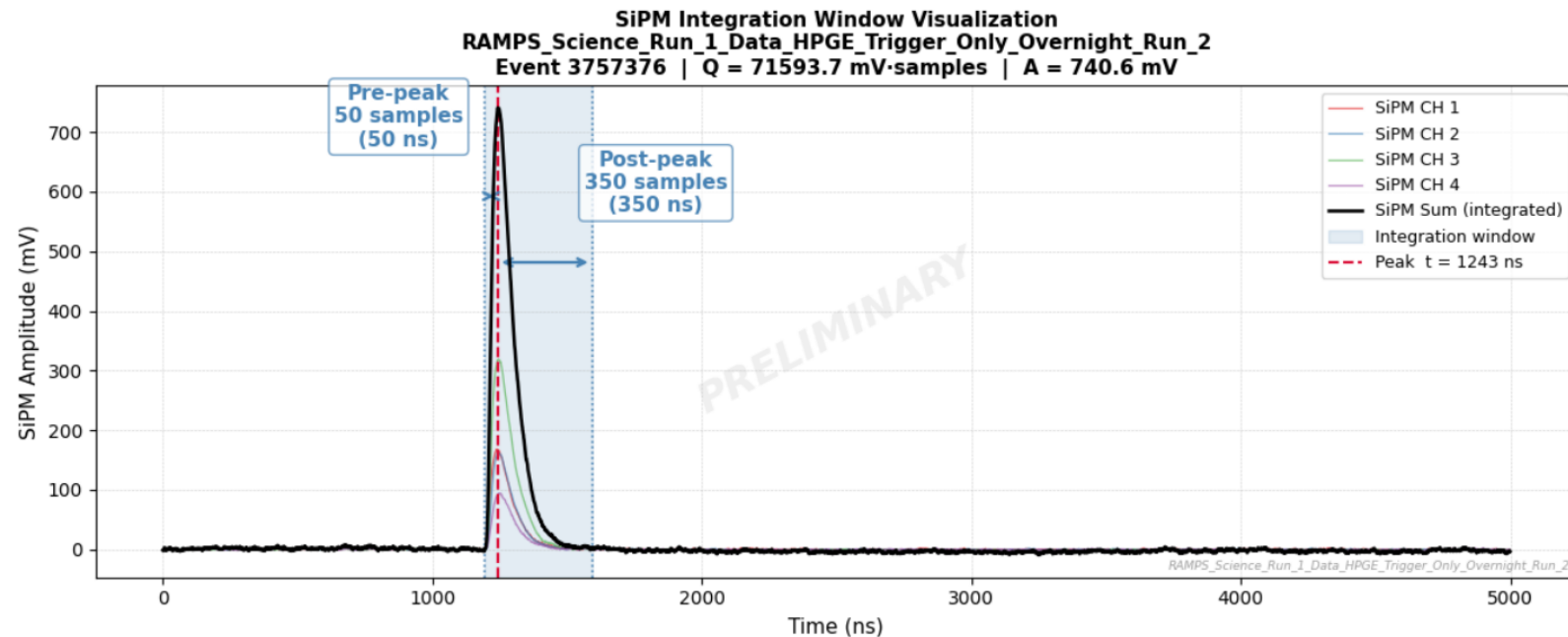
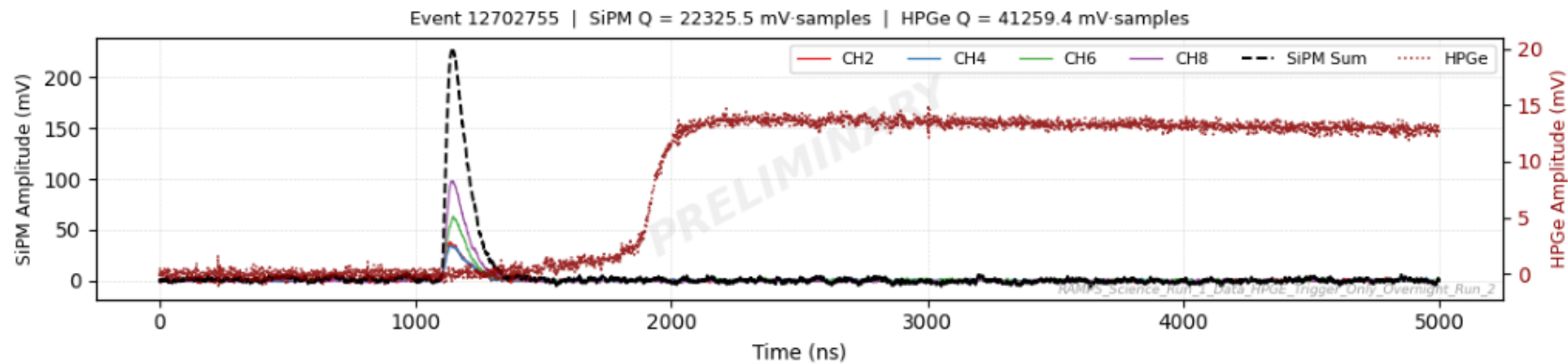


Some of our first pulses!

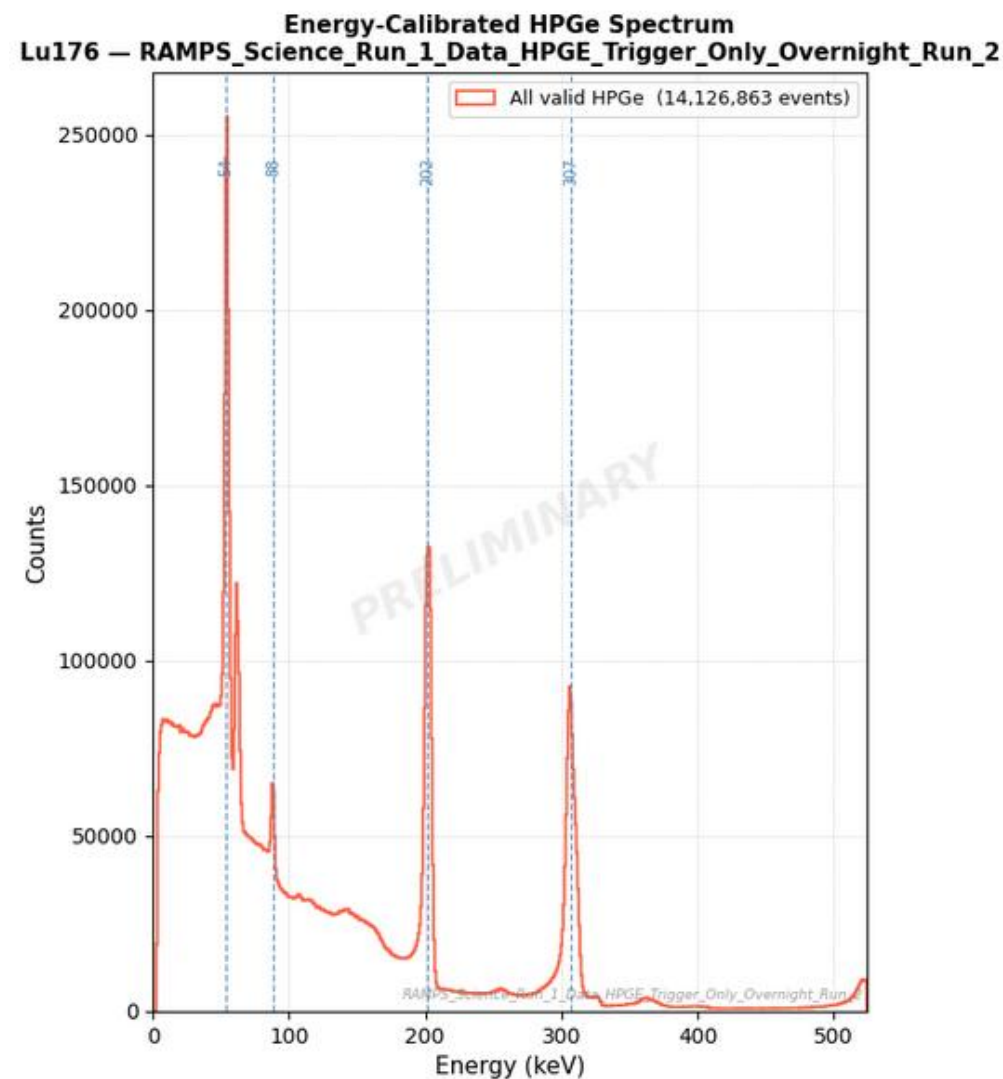
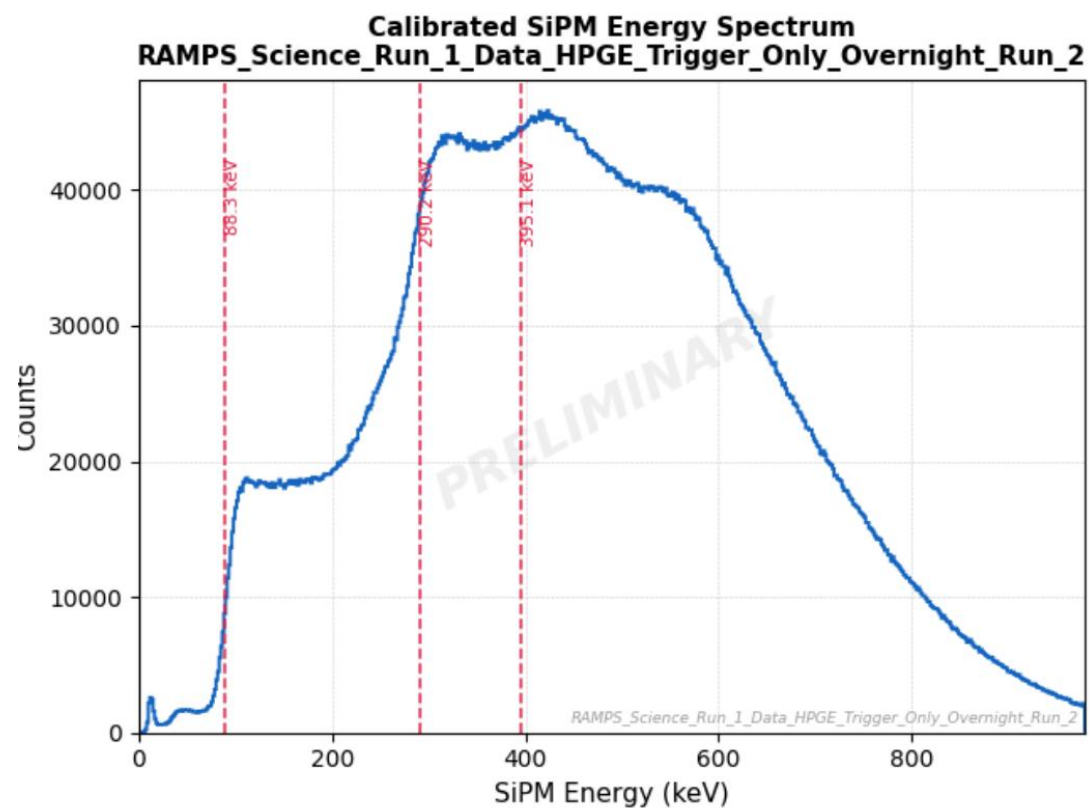


Preliminary Results

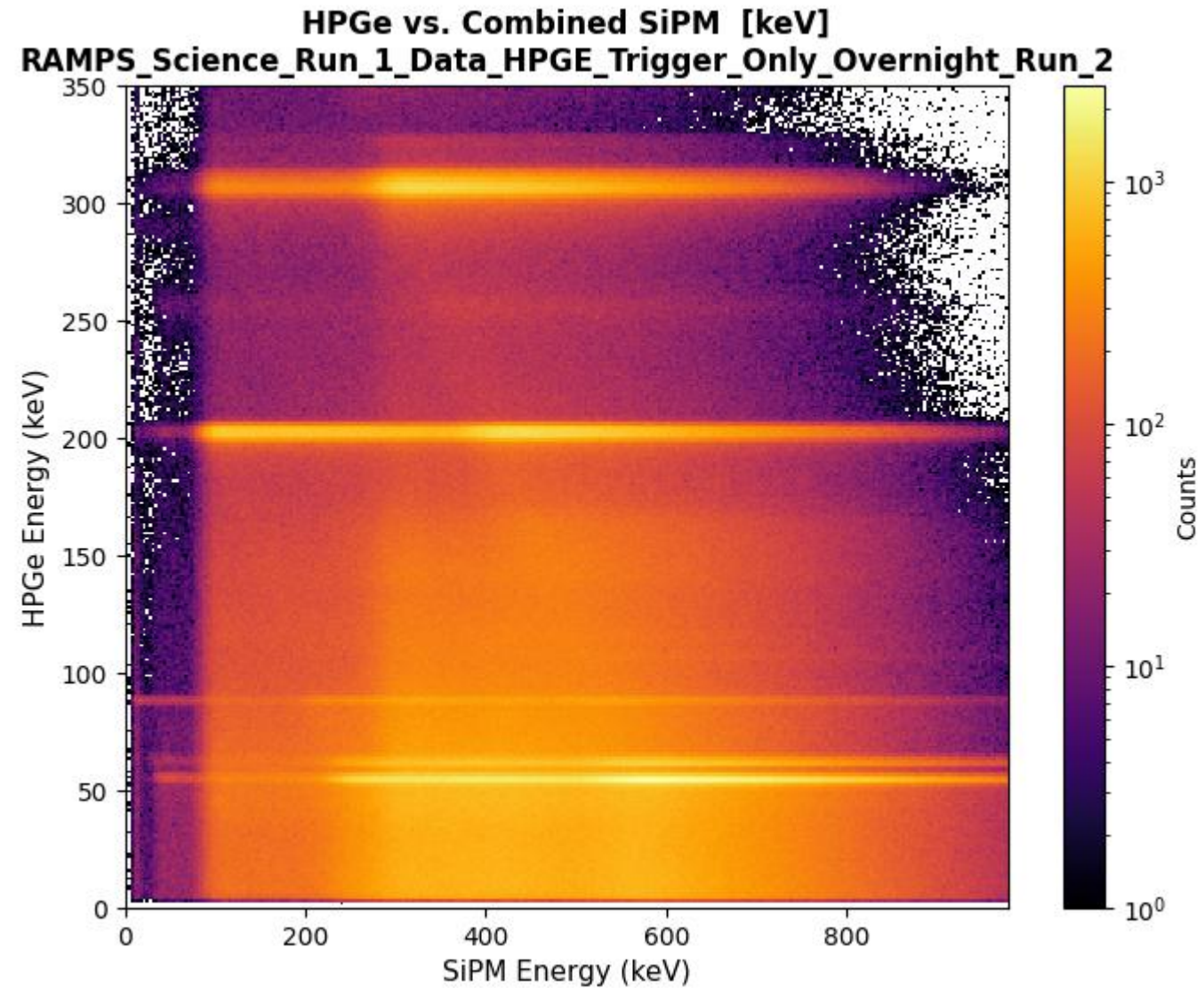
SiPM+ HPGe Waveforms — Bin 70 [22075.3, 22390.7) mV·samples
RAMPS_Science_Run_1_Data_HPGE_Trigger_Only_Overnight_Run_2



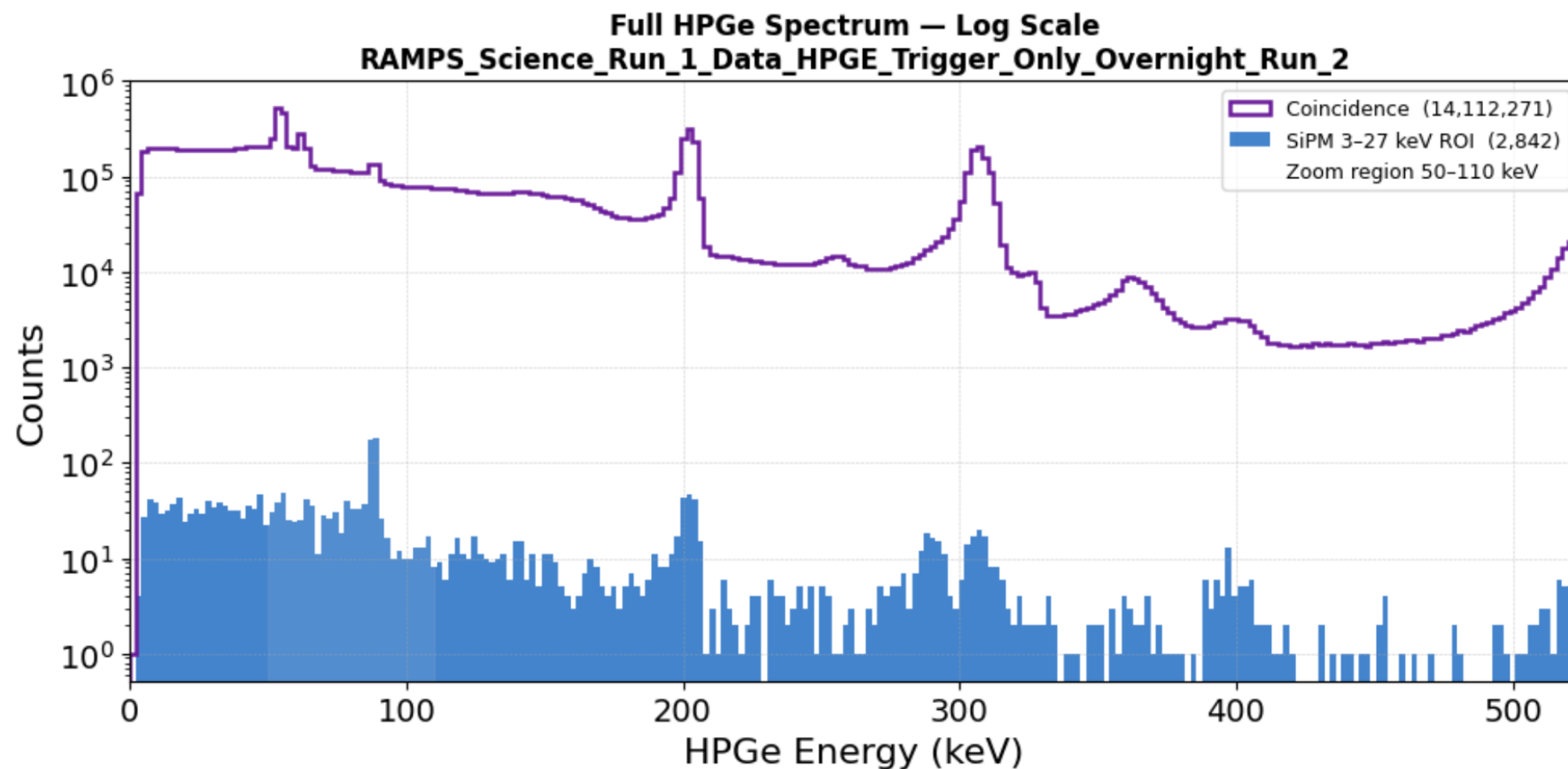
Preliminary Results



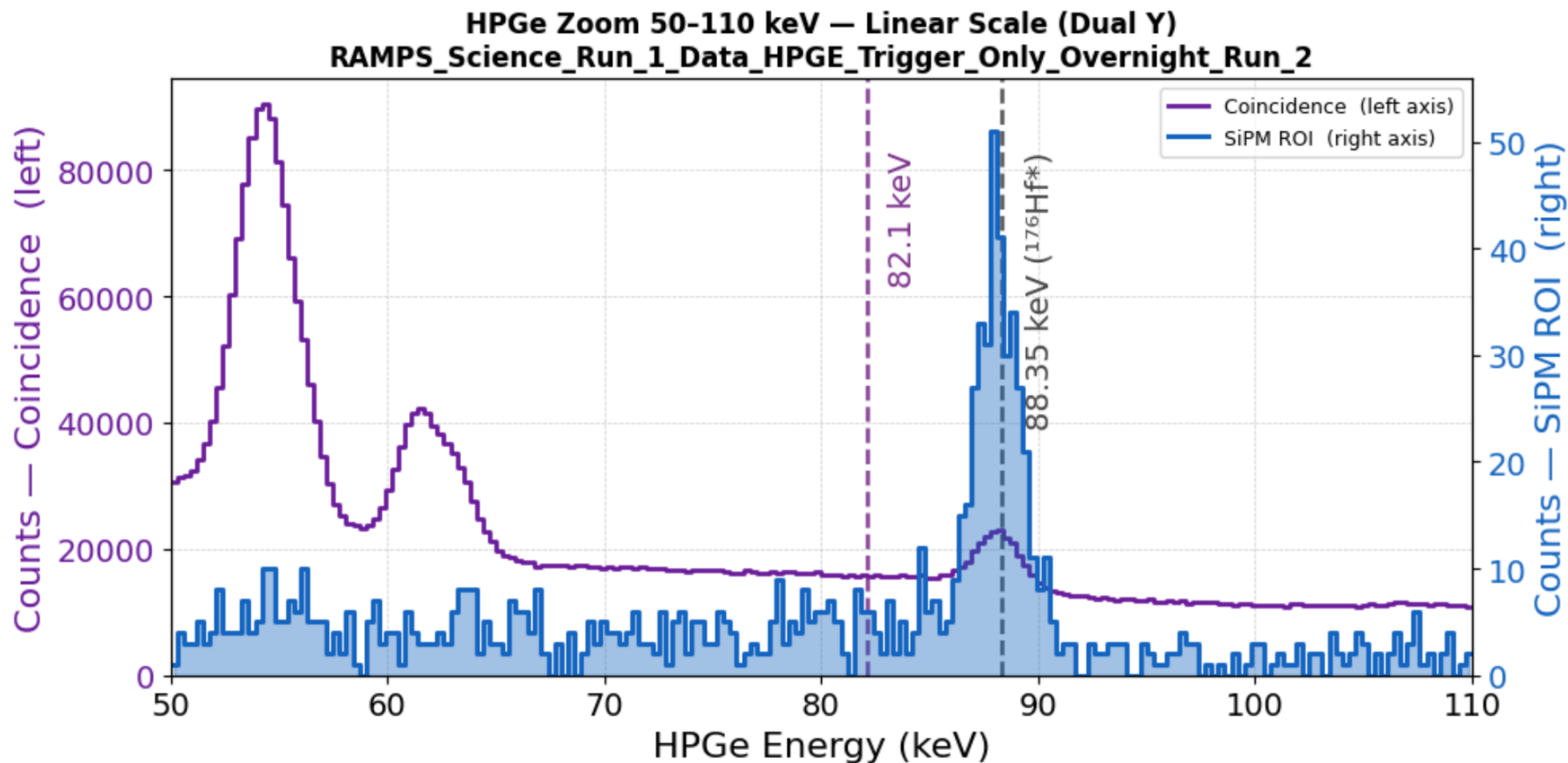
Preliminary Results



Preliminary Results



Preliminary Results

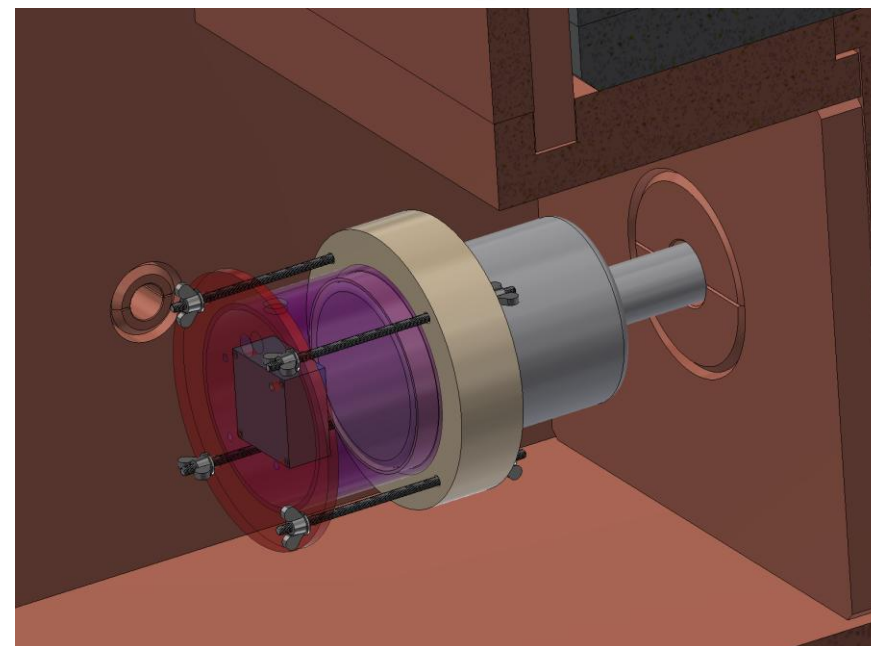
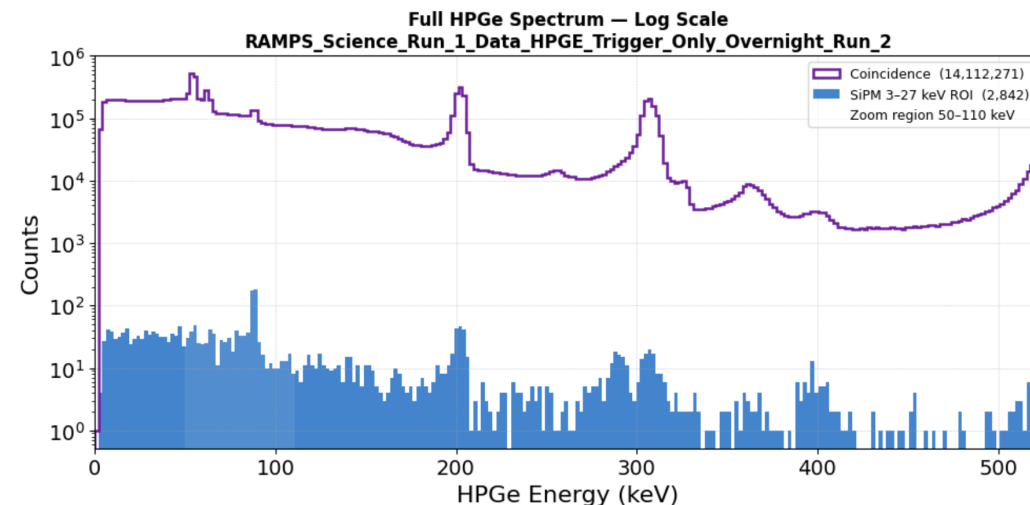


Moving Forward

Further analysis of our data with the 27 keV cut

Improvements for next Science Run:

- New Silicon Photomultiplier
- SiPM holder will be machined
- Left HPGe Detector to be available
- NAS Data storage system



Conclusion

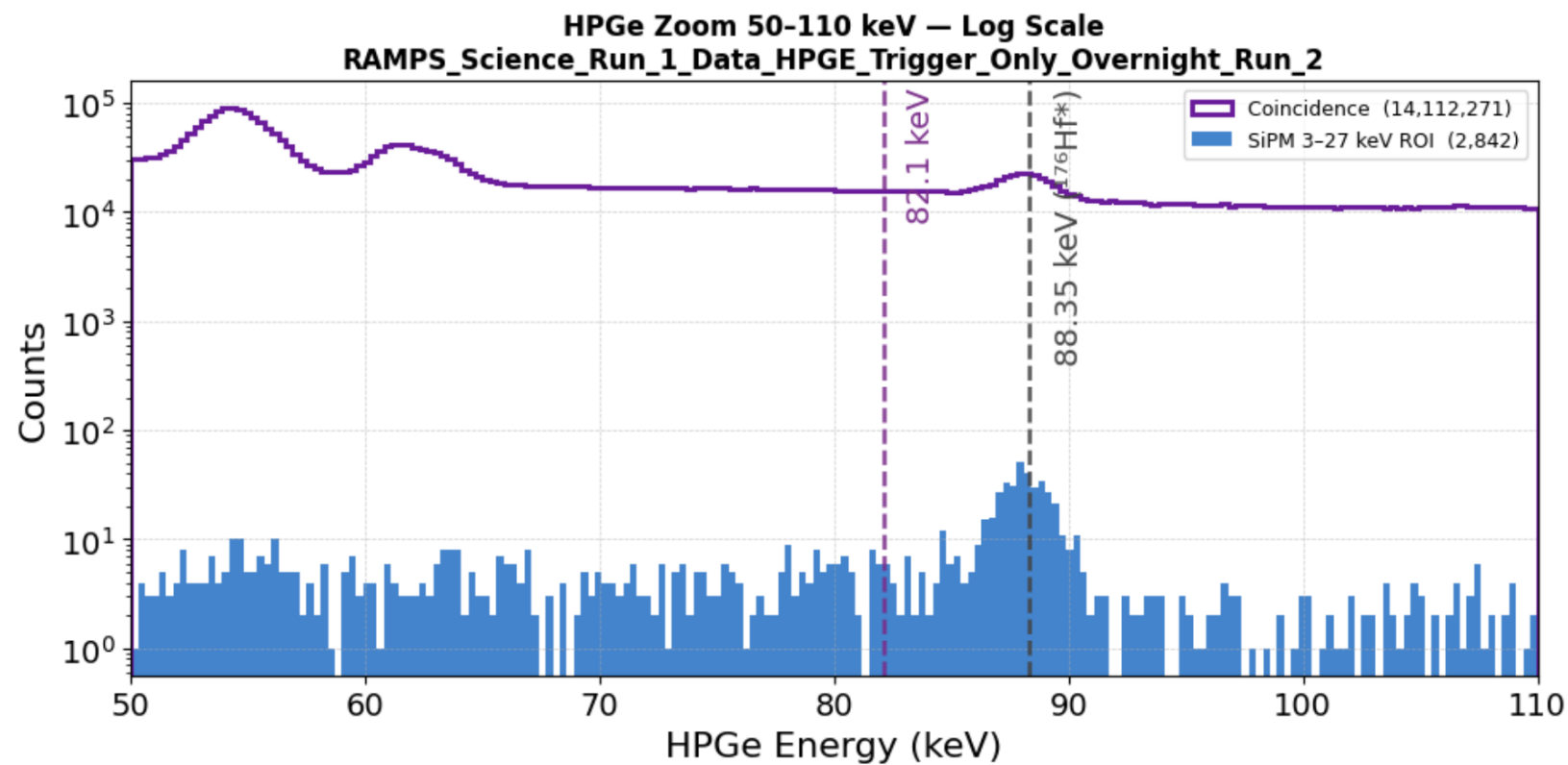
- Very exciting winter term
- Successful Science 1 run
- Great results
- Many improvements are on the way!

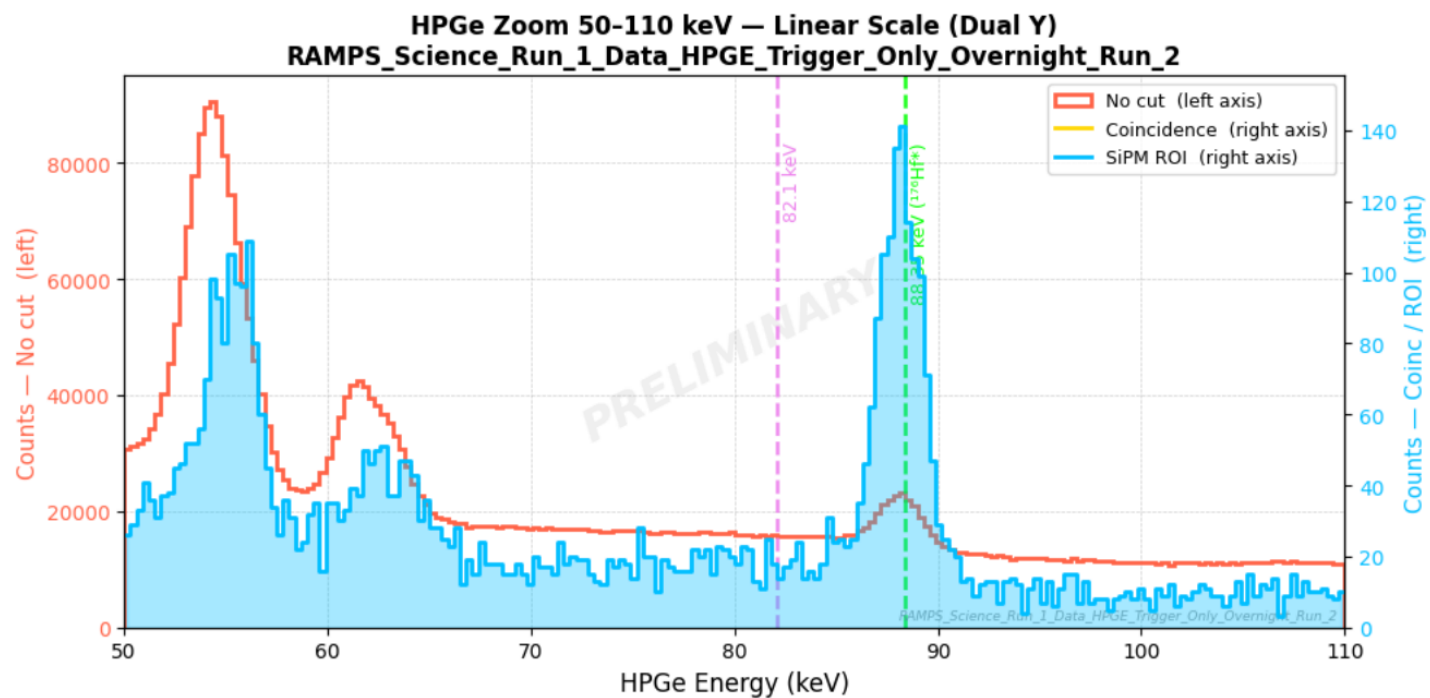
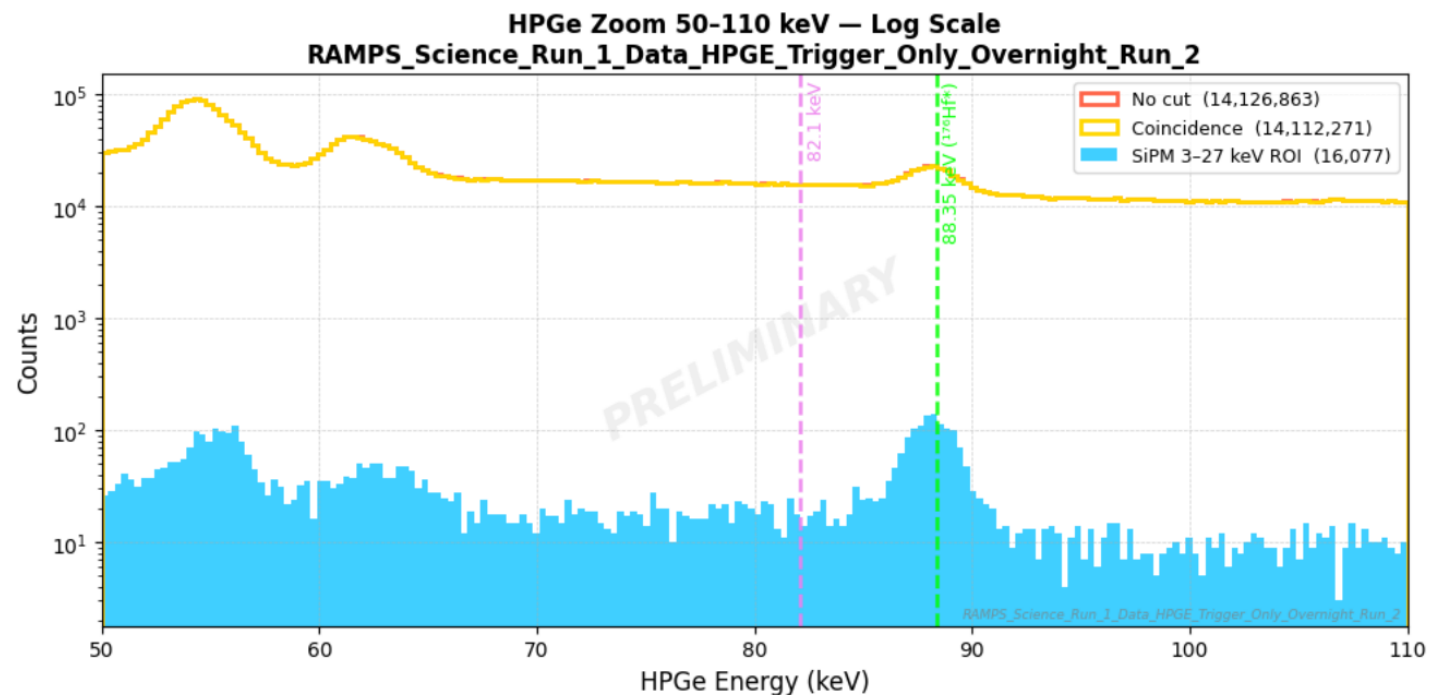


Great Thanks!

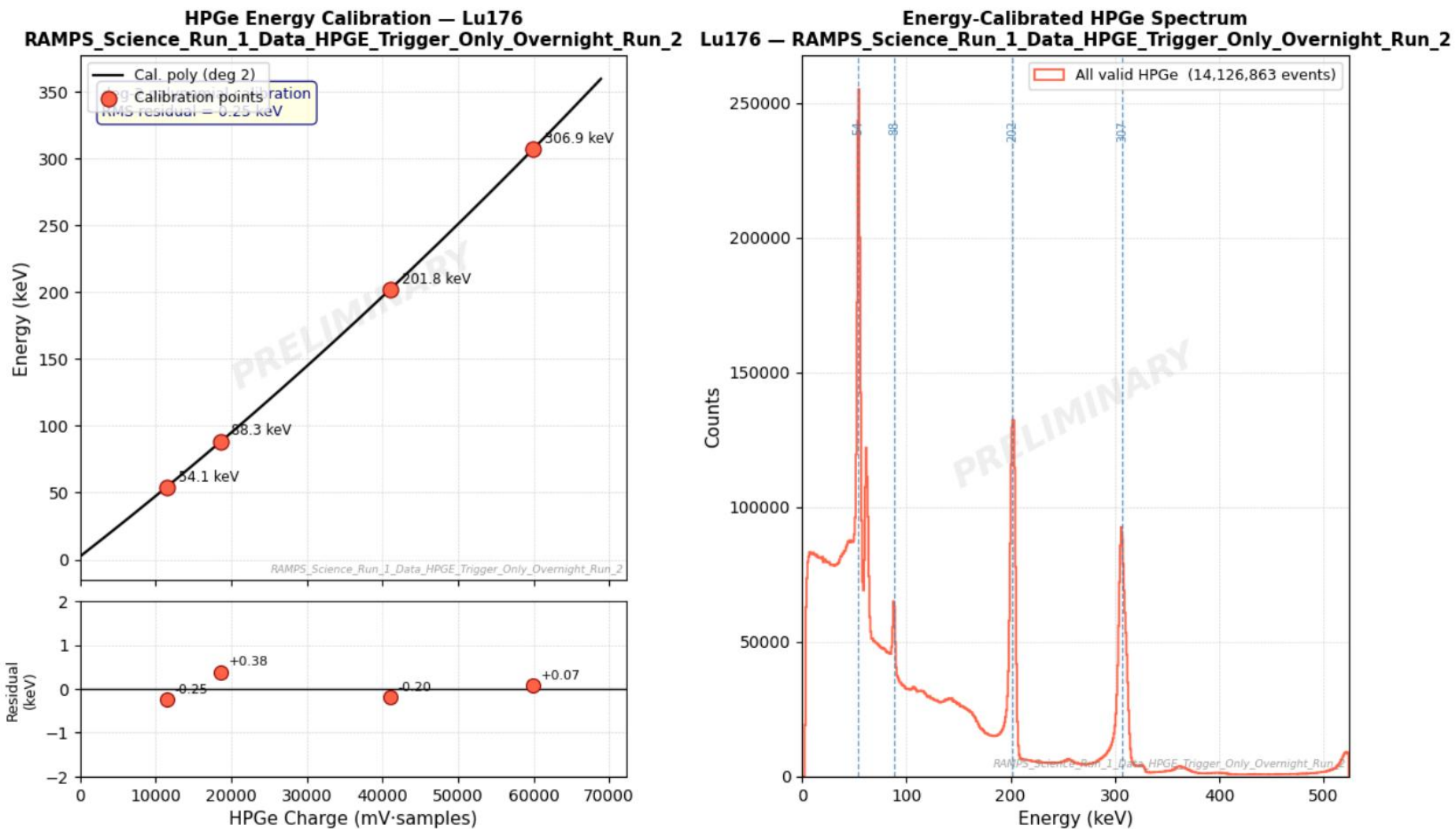
Matthew Stukel, Hector Herrera, Ian Lawson, Steffon Luoma, CAEN Support,
& everybody else who helped me along the way

Questions?

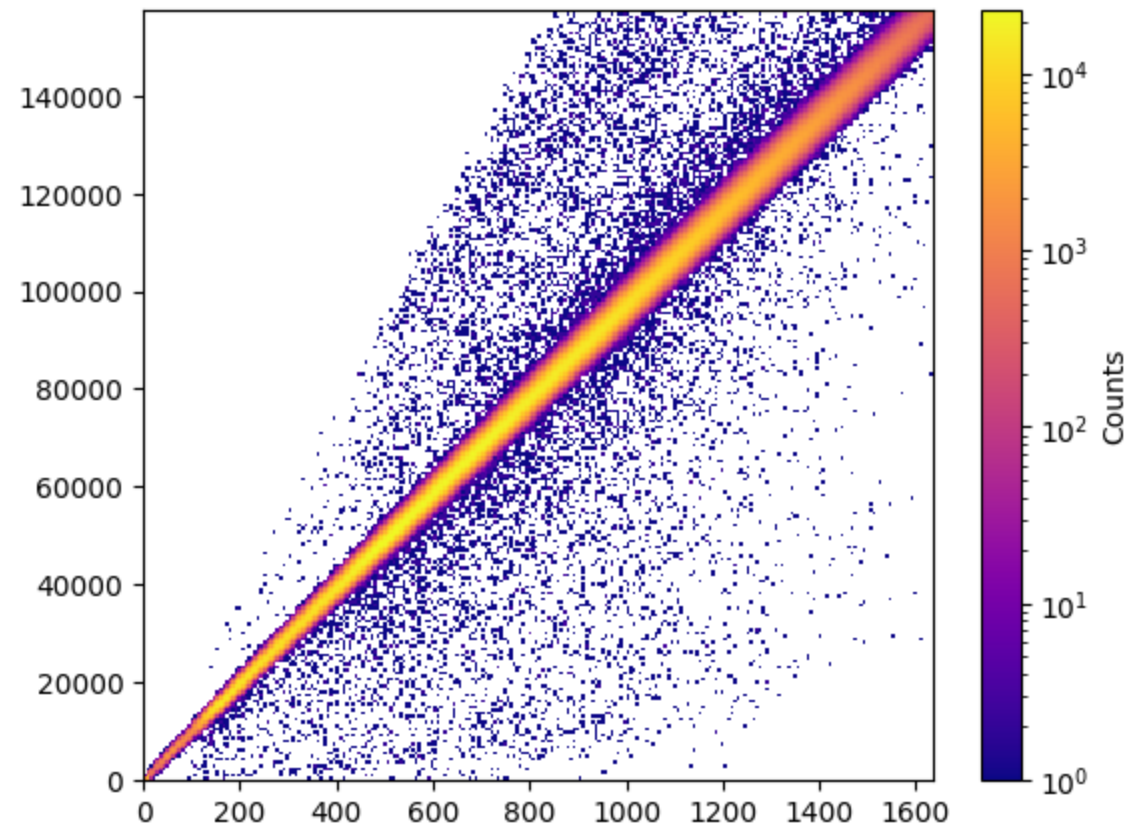
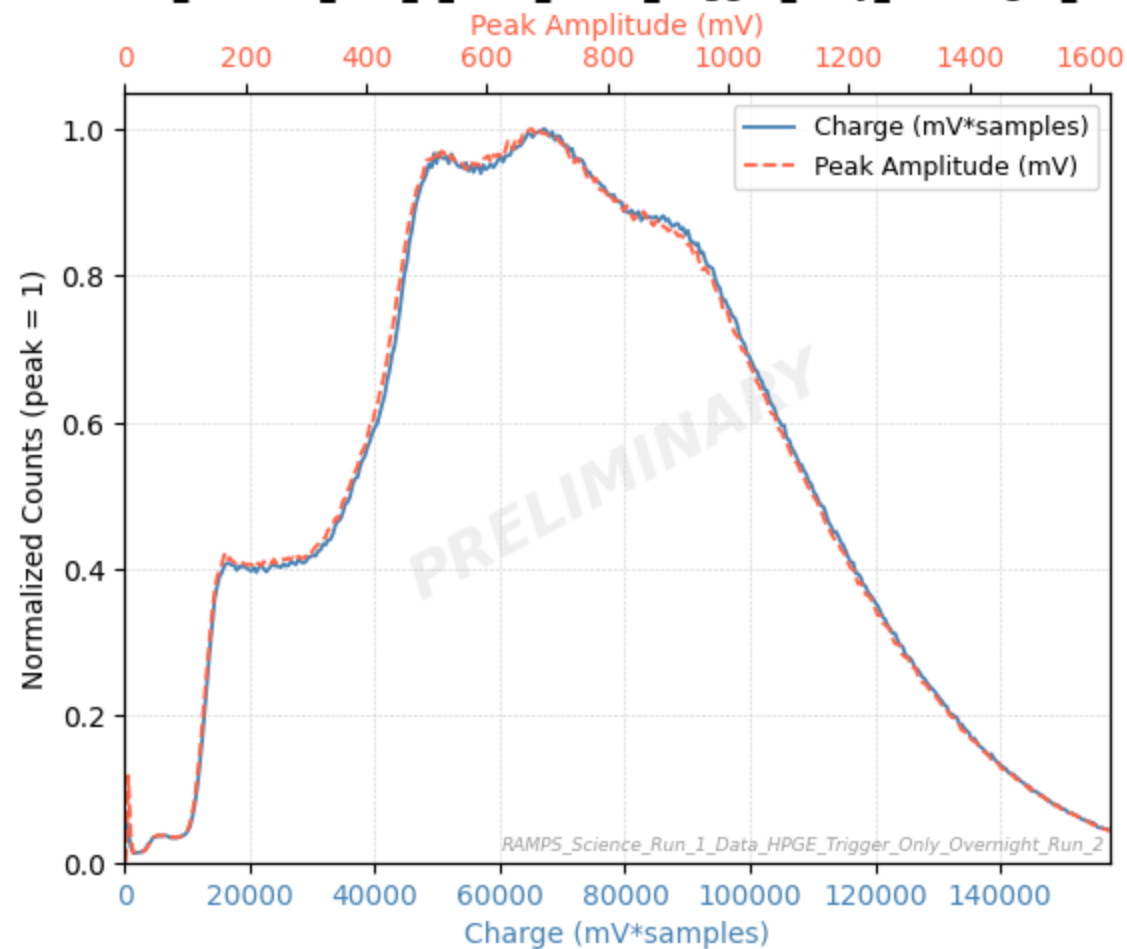




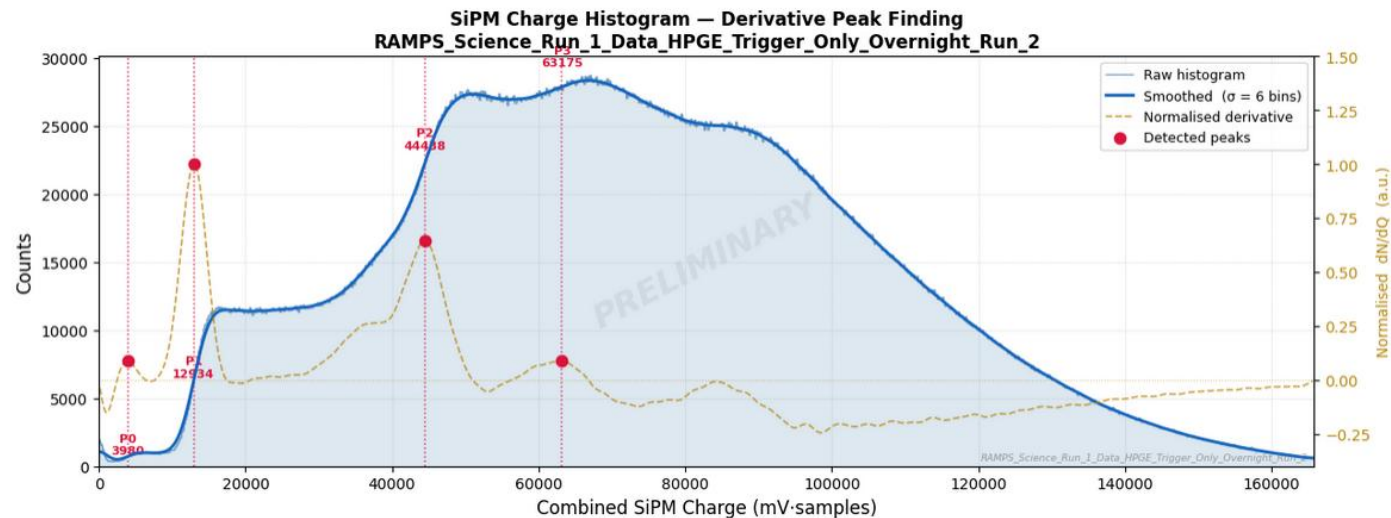
HPGe Energy Calibration Summary — Lu176 — RAMPS_Science_Run_1_Data_HPGE_Trigger_Only_Overnight_Run_2



SiPM Charge vs. Amplitude Spectra (overlaid)
RAMPS_Science_Run_1_Data_HPGE_Trigger_Only_Overnight_Run_2

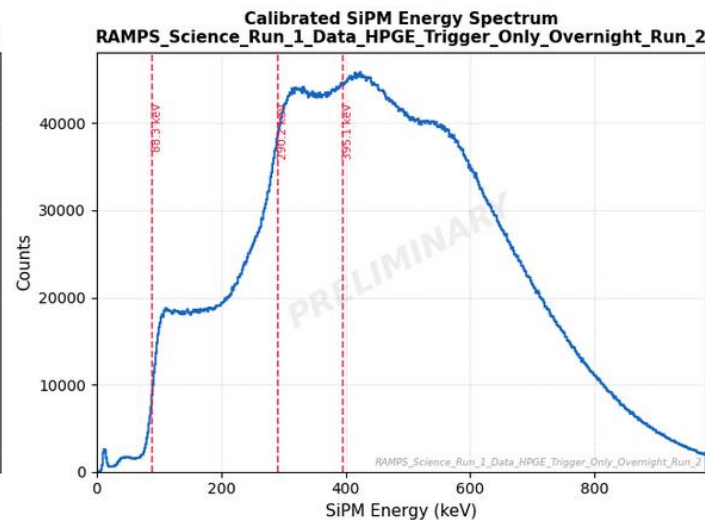
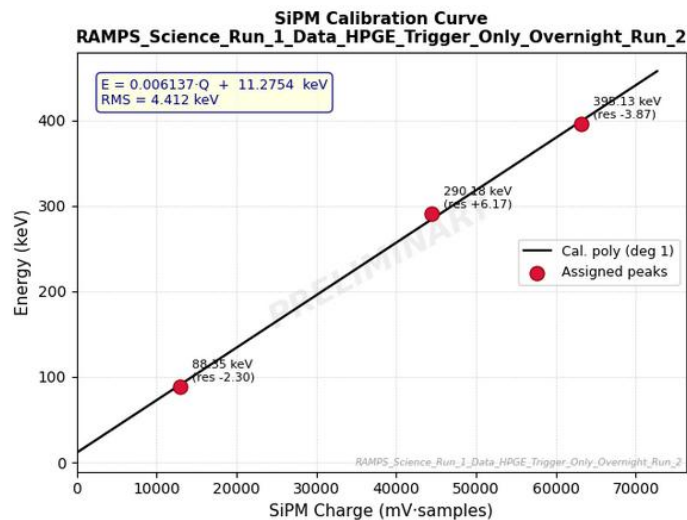


Detected 4 candidate peaks:			
Label	Charge (mV·s)	Deriv. height	Hist. counts
P0	3979.5	35.8	719
P1	12933.5	408.2	6101
P2	44438.3	263.9	22074
P3	63175.3	35.6	27851

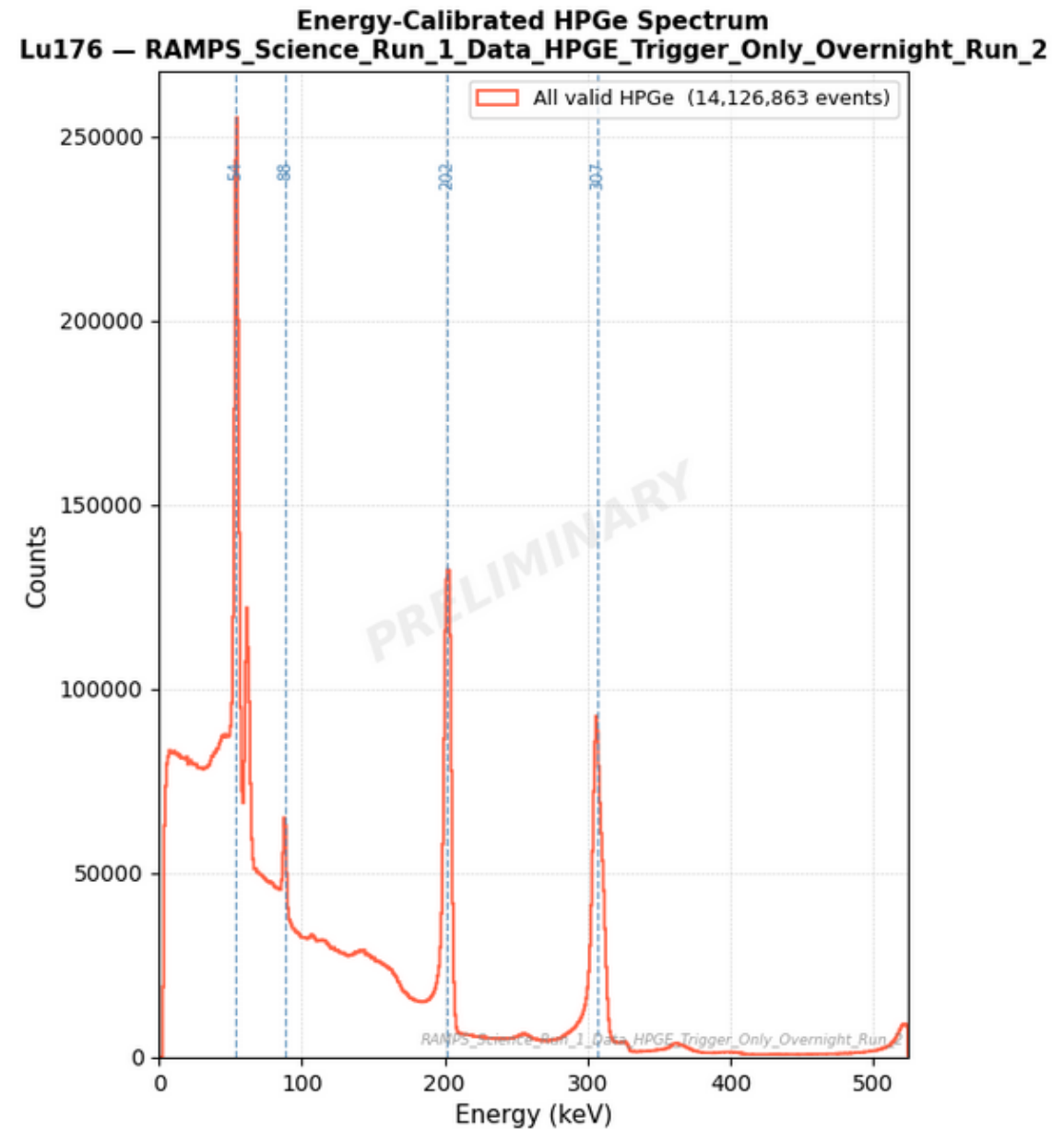
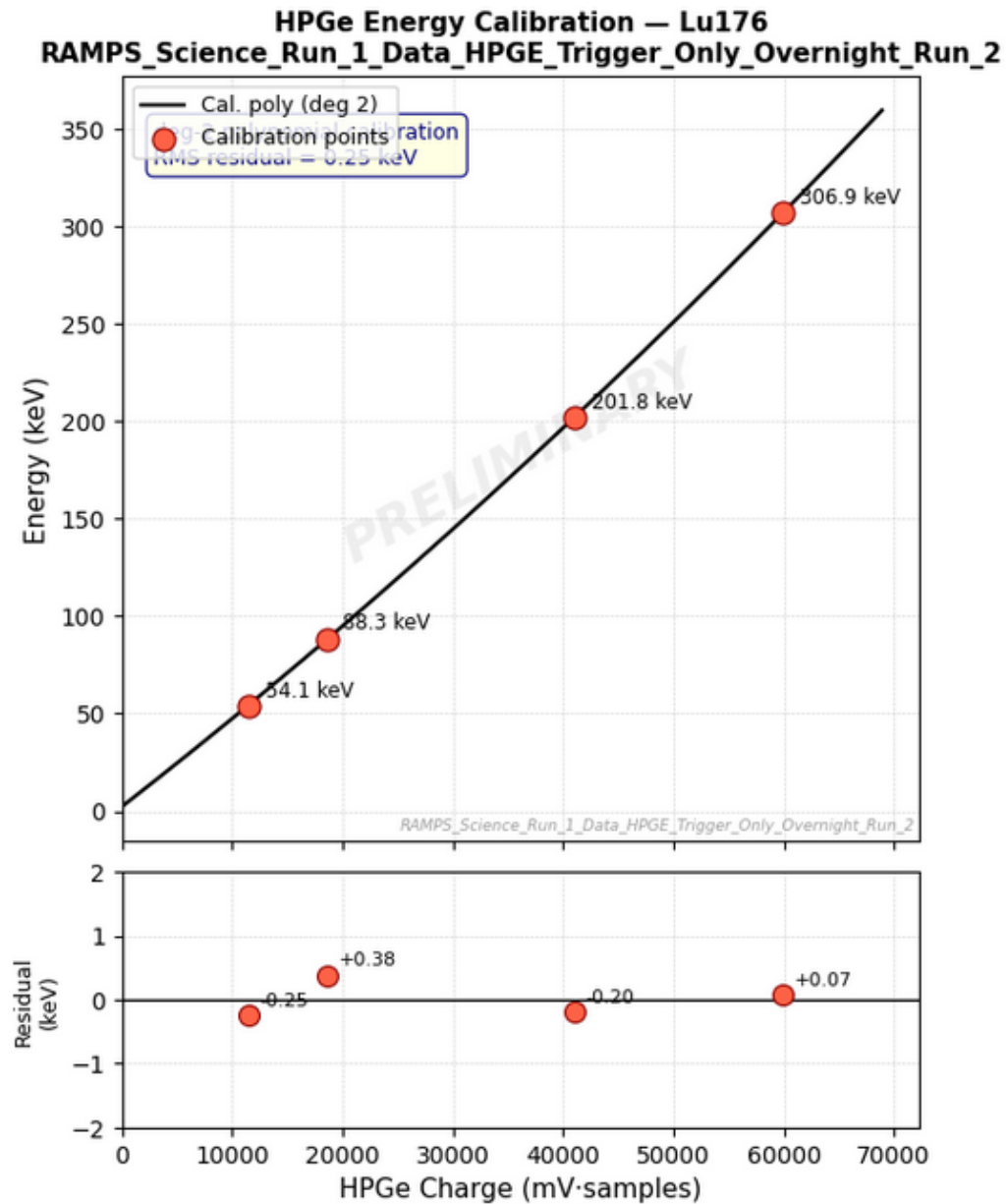


P1: 12933.5 mV·s → 88.35 keV
P2: 44438.3 mV·s → 290.18 keV
P3: 63175.3 mV·s → 395.13 keV

SiPM Calibration: $E = 0.006137 \cdot Q + 11.2754$ keV
Slope : 0.0061373 keV / (mV·s)
Offset : 11.2754 keV
RMS : 4.412 keV



sipm_energy_keV stored - 14,123,787 events above zero.
Energy range: 0.3 - 2749.3 keV



hpge_energy_keV stored — 14,126,863 valid events.

Energy range : 1.8 — 1845.5 keV

B

